Robust-Rail: An Open-Source Platform for Generating, Solving, and Evaluating the Train Unit Shunting Problem with Service Scheduling

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Code https://github.com/Robust-Rail-NL
Demo https://issahanou.github.io/robust-rail-demo/

1 Introduction

The Train Unit Shunting Problem with Service task Scheduling (TUSPwSS) is the logistical challenge of moving trains between stations and shunting yards, where they are parked when not in use and service tasks such as cleaning can be performed [2]. Especially in the Netherlands, where physical space is limited and the available infrastructure is occupied close to capacity, efficient use of the tracks is crucial. Passenger trains in Western Europe are often self-propelled and consist of one or more train units, which are fixed sets of carriages that cannot be split. Train units of the same type can be coupled together to form longer train compositions. The TUSPwSS has four subproblems: 1) the trains must be parked in the shunting yard, 2) they must be routed from the station and within the shunting yard, 3) they must be (de)coupled to form the required train compositions for departure (called the matching subproblem), and 4) the required service tasks (e.g., cleaning or small maintenance) must be performed, at the associated tracks in the shunting yard.

The TUSPwSS has been actively studied, starting from the simpler TUSP [9], also referred to as the Generalized Train Unit Shunting Problem [7]. Various solution methods have been proposed, including mixed-integer linear programming [13], heuristic methods [6], reinforcement learning [8], using a neural network [4], and a local search approach [2]. However, most of these approaches use different datasets that are often proprietary and customized to very specific conditions, making comparison across them challenging and reproducibility difficult.

Therefore, we propose the Robust-Rail framework, which offers an open-source platform for generating instances, computing plans, evaluating and visualizing solutions, and which provides a set of benchmark instances in a standardized format. The local search approach is the state-of-the-art used by railway companies in the Netherlands, and a publicly available version [3] is implemented in our framework. Moreover, we propose a novel approach using automated planning, offering an intuitive approach as a sequential decision-making method. Our planning approach implements a general model for the TUSPwSS in the Planning

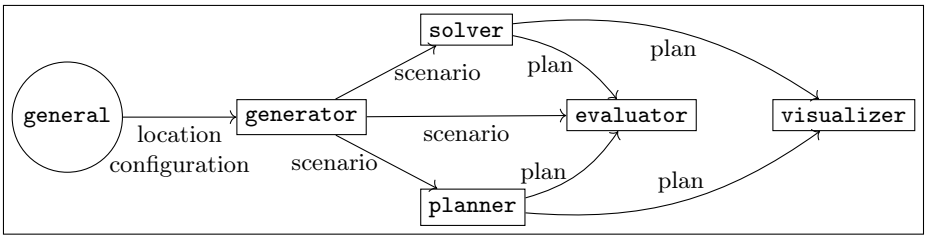


Fig. 1. Overview of the Robust-Rail architecture and workflows.

Domain Definition Language (PDDL) [5] and uses the state-of-the-art numeric planner ENHSP [11] to generate plans. We also implemented a custom heuristic for the TUSPwSS in the SymbolicPlanners framework [14], in which we impose an intuitive priority on actions to prune the search space. Finally, we use a previously developed simulator, TORS [10], to evaluate plans.

2 Overview

Our framework contains six components (Figure 1). First, the `robust-rail-general` repository contains benchmark instances for different locations, a set of example plans, and several pipelines to run the precompiled modules for the different components. The first pipeline component is the `robust-rail-generator`, which generates the actual instances given a location and a scenario configuration (which can be fully specified or randomized). Next, we have the `robust-rail-solver`, which includes the codebase for the local search solver used to generate a plan for an instance. Alternatively, the pipeline can use the `robust-rail-planner` component to generate a plan. Given a plan and the associated instance, the `robust-rail-evaluator` checks whether the plan is valid and computes the resulting delay. Finally, the `robust-rail-visualizer` visualizes a plan on the location representation and shows the sequential steps either iteratively or as a whole. The repository is open to implement other solvers as well, such as a first attempt at Lagrangian Relaxation [12].

3 Conclusion

Robust-Rail is an open-source framework for researching the Train Unit Shunting Problem with Service task Scheduling by generating instances, solving them, evaluating plans, and visualizing them. Our standardized format also allows for the addition of other locations or even similar problem statements of similar transportation problems. Future work will focus on expanding the framework not only to the TUSPwSS but also to encompass the entire railway hub planning problem [1]. The repository is open to adding more solution methods for this interesting but complex real-world problem.

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